Pat Lank

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POSITIONS

- **Università degli Studi di Milano, Italy** Fall 2024 | Present
  - Postdoctoral research fellow
  - Supervisor: Amnon Neeman
- **Simons-Laufer Mathematical Sciences Institute, USA** April 2024
  - Research associate

EDUCATION

- **University of South Carolina, USA** January 2021 to May 2024
  - PhD in Mathematics
  - Advisor: Matthew Ballard
- **University of New Mexico, USA** August 2017 to December 2020
  - M.Sc. in Mathematics
  - Advisor: Alexandru Buium
- **University of Massachusetts Lowell, USA** August 2015 to August 2017
  - B.Sc. in Mathematics

RESEARCH

Interests. Algebraic geometry, especially towards singularity theory, using methods from derived categories and homological algebra.

Published & accepted.

(1) *Compact approximation and descent for algebraic stacks*
- with Jack Hall, Fei Peng, Alicia Lamarche
- `arXiv`, accepted to **Forum Math. Sigma**

(2) *Measuring birational derived splinters*
- with Timothy De Deyn, Kabeer Manali Rahul, Sridhar Venkatesh
- `arXiv`, **Bull. Lond. Math. Soc.**

(3) *Integral transforms on singularity categories for Noetherian schemes*
- with Uttaran Dutta, Kabeer Manali Rahul
- `arXiv`, accepted to **Michigan Math. J.**

(4) *Measuring rationality of Schwede–Takagi pairs*
- with Peter McDonald, Sridhar Venkatesh
- `arXiv`, **J. Pure Appl. Algebra**

(5) *Triangulated characterizations of singularities*
- with Sridhar Venkatesh
- `arXiv`, **Nagoya Math. J.**

(6) *Closedness of the singular locus and generation for derived categories*
- with Souvik Dey
- `arXiv`, **J. Algebra**

(7) *Dévissage for generation in derived categories*
- with Souvik Dey
- `arXiv`, **Proc. Amer. Math. Soc.**

(8) *Approximation by perfect complexes detects Rouquier dimension*

- with Noah Olander
- arXiv, **Mosc. Math. J.**

(9) *Preservation for generation along the structure morphism of coherent algebras over a scheme*
- with Anirban Bhaduri, Souvik Dey
- arXiv, **Bull. Lond. Math. Soc.**

(10) *Descent conditions for generation in derived categories*
- arXiv, **J. Pure Appl. Algebra**

(11) *Strong generation for module categories*
- with Souvik Dey, Ryo Takahashi
- arXiv, **J. Pure Appl. Algebra**

(12) *High Frobenius pushforwards generate the bounded derived category*
- with Matthew Ballard, Srikanth Iyengar, Alapan Mukhopadhyay, Josh Pollitz
- arXiv, **Forum Math. Sigma**

Under revision.

(1) *Fppf descent and localizing subcategories for algebraic spaces*
- arXiv
- *European Math. J.*

(2) *Descent and generation for noncommutative coherent algebras over schemes*
- with Timothy De Deyn, Kabeer Manali Rahul, arXiv
- *J. Lond. Math. Soc.*

Selected preprints. A complete list of preprints is available on my webpage; see also my arXiv author page.

(1) *Germwise singular Fourier–Mukai partnerships*
- arXiv

(2) *Gluing weak approximability along recollements*
- arXiv

(3) *A blowup criterion for regularity*
- arXiv

(4) *Fourier–Mukai loci are open and base change*
- with Elías Guisado Villalgordo
- arXiv

(5) *Proxy smallness meets t-structures*
- with Michal Hrbek, Giovanna Le Gros, Sergio Pavon
- arXiv

(6) *Remarks on diagonal dimension for algebraic stacks*
- with Fei Peng
- arXiv

(7) *Dualizing complexes and t-structures for algebraic spaces*
- arXiv

(8) *Fiberwise criteria for Fourier–Mukai equivalences*
- with Elías Guisado Villalgordo, Kabeer Manali Rahul, Nebojsa Pavic
- arXiv

(9) *Nonexistence of singly compactly generated t-structures for schemes*
- with Anirban Bhaduri, Timothy De Deyn, Michal Hrbek, Kabeer Manali Rahul
- arXiv

(10) *A note on quasi-perfect morphisms*

- with Timothy De Deyn, Kabeer Manali Rahul
- arXiv

(11) *Simple criteria for higher rational singularities*
- with Sándor Kovács, Sridhar Venkatesh
- arXiv

(12) *Perfectly generated t-structures for algebraic spaces*
- with Michal Hrbek, Simone Pizzirani
- arXiv

INVITED TALKS & LECTURES

Invited talks.

(34)	Universität Graz	TBA
(33)	Oxford Algebraic Geometry Days	September 2026
(32)	International Conferences on Representations of Algebras (ICRA)	June 2026
(31)	Univerzita Karlova	June 2026
(30)	University of Pennsylvania	April 2026
(29)	AMS Special Session, Tulane University	March 2026
(28)	Università di Pavia	March 2026
(27)	Max Planck Institute for Mathematics	February 2026
(26)	Conference at University of Genova	June 2025
(25)	Università degli Studi di Milano	April 2025
(24)	University of Glasgow	March 2025
(23)	Univerzita Karlova	February 2025
(22)	Universität Hamburg	January 2025
(21)	AGNES, Dartmouth University	November 2024
(20)	Purdue University	October 2024
(19)	AMS Special Session, Howard University	April 2024
(18)	COMA/NAG, Simons-Laufer Mathematical Sciences Institute	March 2024
(17)	AMS contributed paper session, Joint Mathematics Meetings	January 2024
(16)	University of Georgia	October 2023
(15)	New directions in group theory and triangulated categories	May 2023
(14)	Georgia algebraic geometry symposium (GAGS), University of Georgia	May 2023
(13)	Categorical methods in moduli theory, University of Pennsylvania	April 2023
(12)	AMS Special Session, Spring Central Sectional	April 2023
(11)	AMS Special Session, Southeast Sectional	March 2023
(10)	University of Utah	September 2022
(9)	Algebraic geometry & singularity theory, University of Washington	June 2022
(8)	Commutative algebra regional expository seminar	April 2022
(7)	University of South Carolina	February 2022
(6)	Commutative algebra regional expository seminar	October 2021
(5)	Algebraic geometry number theory, University of South Carolina	March 2021
(4)	University of New Mexico	November 2019
(3)	University of New Mexico	December 2018
(2)	Women in mathematics in New England, Smith College	September 2016
(1)	MAA northeast spring section meeting, University of New England	June 2016

Lectures.

(3)	Summer school on mirror symmetry, University of Zaragoza for RGAS	June 2026

(2) Basque Center for Applied Mathematics, Spain — September 2025
(1) Syzygies & mirror symmetry, Amer. Inst. of Math. — September 2023

TEACHING

University of South Carolina.

Instructor of Record. Completely responsible for lectures, office hours, exams, homework assignments, and grades.

- `MATH 122 - Business calculus` — Spring 2024
- MATH 111i - Intensive Basic College Mathematics — Fall 2023
- MATH 174 - Discrete structures for computer science — Spring 2023
- MATH 111 - Basic college mathematics — Fall 2022
- MATH 241 - Calculus III — Summer 2022
- MATH 122 - Business calculus — Spring 2022
- MATH 111 - Basic college mathematics — Fall 2021 (overload)
- MATH 115 - Precalculus — Fall 2021
- MATH 241 - Calculus III — Summer 2021
- MATH 122 - Business calculus — Spring 2021

University of New Mexico.

Instructor of Record. Completely responsible for lectures, office hours, exams, homework assignments, and grades.

- MATH 180 - Calculus I — Summer 2020
- MATH 103 - Intermediate algebra III — Fall 2018
- MATH 102 - Intermediate algebra II — Fall 2018
- MATH 101 - Intermediate algebra I — Fall 2018
- MATH 121 - College algebra — Fall 2017

Graduate Teaching Assistant. Completely responsible for grading exams and homework assignments.

- MATH 521 - Abstract algebra (graduate course) — Spring 2020
- MATH 319 - Number theory — Spring 2020
- MATH 327 - Discrete structures — Spring 2019
- MATH 322 - Modern algebra — Spring 2019
- MATH 321 - Linear algebra with applications — Fall 2019

GRADUATE SUPERVISION & MENTORSHIP

- Elías Guisado Villalgordo, PhD Student at Basque Center for Applied Mathematics (BCAM)
  - Co-advisor with Ilya Smirnov
  - Expected 2028

GRANTS & AWARDS

- NSF MSPRF (recommended for award; not funded) — Spring 2026
- Marie Skłlodowska-Curie Seal of Excellence — Spring 2026
- NSF MSPRF (recommended for award; not funded) — Spring 2025
- Outstanding graduate student award in mathematics (UofSC) — Spring 2024
- Teaching award from Student Disability Resource Center (UofSC) — Fall 2023

- AMS Mathematical Research Communities — Summer 2023
- AMS graduate student sectional travel grant — Spring 2023
- AMS-NSF-Simons-ICM travel grant — Spring 2022

SERVICE

Referee & reviewing service.

- Bull. Lond. Math. Soc.
- Appl. Categ. Struct.
- Rend. Sem. Mat. Univ. Padova
- zbMATH Open

Committee.

- Math 111 textbook committee for UofSC — Spring 2024
- Mathematics Tutoring Center coordinator — Summer 2023
- Graduate student panel committee — Spring 2022
- Graduate student panel committee — Summer 2021
- Proctor for UNM PNM statewide mathematics exam — Spring 2019
- Proctor for UNM PNM statewide mathematics exam — Fall 2017
- City-wide concert and fundraiser, Nashua NH Soup Kitchen — Fall 2011

Organization.

- `Mini workshop`: Derived categories in algebra, geometry, & topology — June 2025
- Algebraic geometry `seminar` (UniMi) — Spring 2025 to January 2026
- D.A.N.C.E. seminar `(online seminar)` — January 2025 to June 2026
- JMM: `Special Session` on derived categories, arithmetic & geometry — January 2024
- `Graduate colloquium` (UofSC) — Fall 2021 to Spring 2023
- Algebraic geometry and commutative algebra `seminar` (UofSC) — Fall 2021 to Spring 2023

Misc.

- Over ten years of experience in retail and food service, including supervisory duties such as training employees, coordinating high-volume operations, inventory auditing, and opening/closing procedures.